

Estimate Height Weight and Body Mass Index(BMI)From Image Face Using Machine Learning

ABSTRACT

This study investigates the use of machine learning to estimate height, weight, and body mass index (BMI) from facial images. By employing Regression models, convolutional neural networks (CNNs), the research aims to develop a model that can infer these biometric parameters based on facial features. The model is trained on a large dataset of facial images with corresponding height and weight data, exploring the correlation between facial morphology and body metrics. The ultimate objective is to create a non-invasive tool for BMI estimation that can be integrated into health monitoring systems and personalized health applications. Human body photographs contain encodings for gender, weight, and other biometric data in addition to pupil colour. When considering all of this information, body weight is a reliable indicator of health problems. Body measurements including height, weight and body mass index (BMI) are important because they can be used for a variety of purposes like surveillance, re-identification, picture retrieval systems, and healthcare. The majority of earlier research on automated height, weight, and BMI measurement has made use of 3D whole body videos and 2D and full body 2D pictures. A person's weight in proportion to their height is gauged by their BMI. It serves more as a gauge for calculating total body fat. Because it is frequently used to evaluate health issues, BMI is important. Our chances of living a longer, healthier life are said to be increased by having a healthy body mass index(BMI).